Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: M.Tech Structural Engineering
Name of the Course: Soil-Structure Interaction
Time: 90 minutes
Note:

Semester: III
Course Code: MSE315
Max. Marks: 50

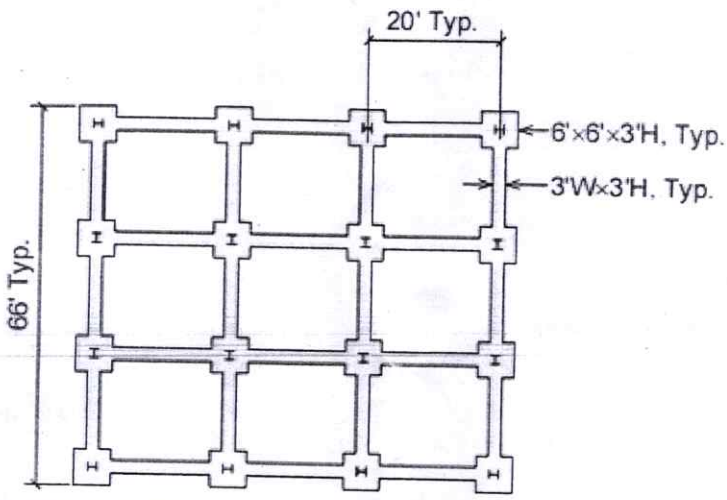
(i) ***Answer all the questions*** by choosing ***any one of the sub-question***
(ii) ***Each question carries 10 marks***

Q1		
	10 marks	**COs**
(a)	Discuss in detail the sub-structure approach and direct analysis approach for SSI analysis.	**CO-1**
	OR	
(b)	Discuss the following terminology normally used in SSI (i) Spatial variable motion (ii) Kinematic Interaction	**CO-1**

Q2		
	10 marks	**COs**
(a)	Discuss how "High Structure to Soil Stiffness Ratios will Lengthen Period and Change Design Forces.	**CO-1**
	OR	
(b)	As per ASCE/SEI 7-16, How the following information are calculated or obtained to take into account SSI effect (i) Effective profile depth (ii) Effective depth for foundation rotation	**CO-1**

Q3		
	10 marks	**COs**
(a)	Give examples of "Building Base Configuration" with diagram showing different degrees of connectivity.	**CO-2**
	OR	
(b)	Discuss the step by step procedure to account for base slab averaging account for considering SSI Effect	**CO-2**

Q4		
	10 marks	**COs**
(a)	For the given building plan, calculate the ratio of footing area to the foot print area	**CO-2**

		
	OR	
(b)	Discuss the step by step procedure involved while taking into account the embedment effect on SSI	**CO-2**

Q5	**10 marks**	**COs**
(a)	Discuss how Foundation Rocking Impacts Superstructure Behavior?	**CO-1**
	OR	
(b)	Discuss How Substantial Foundation Embedment Reduces Design Forces?	**CO-1**